Identifying Skill Gaps Between University Curricula and Industry Job Requirements

MASTER'S DISSERTATION PRESENTATION



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Research Problem

IMPORTANCE OF CURRICULUM–INDUSTRY ALIGNMENT

Understanding the gap between academic curricula and industry needs is crucial for ensuring that Data Science graduates develop skills relevant to the evolving workforce.

- Ensures graduates are equipped with relevant and industry-demanded skills.
- Addresses skill shortages and mismatches in the workforce.
- Enhances the employability and career readiness of graduates.
- Supports continuous improvement and updating of university curricula.



Why This Study Matters

A mismatch between what universities teach and what industry requires can create skill gaps that affect graduate employability and workforce preparedness.

Research Objectives and Questions



KEY ANALYTICAL FOCUS

The main objective is to identify skill gaps between academic curricula and industry requirements.

The study examines where industry skill demand and curriculum representation converge, diverge, and reveal gaps in curriculum coverage.

Research Question

To what extent do university Data Science curricula align with the skills and priorities represented in industry requirements, and where are the major gaps in curriculum coverage?

Eight Analytical Questions

1. Which skills are most frequently represented in industry and curriculum documents?
2. How do industry and curriculum priorities differ across skill categories?
3. How closely do comparable skill prevalences align?
4. Where are the largest industry–curriculum skill gaps?
5. How much of industry skill demand is represented in curricula?
6. How well are selected emerging technologies represented?
7. Do industry and curriculum skill rankings differ substantially?
8. Is curriculum coverage associated with industry skill categories?

Data Sources and Study Design Overview

DATASETS AND METHODOLOGY

This study utilizes industry and curriculum datasets to compare skill requirements, curriculum representation and skill gaps across 14 standardized skill categories.



Industry Dataset

130 job-description documents
81 standardized industry skills
Prevalence denominator: 130

Curriculum Dataset

80 curriculum documents
79 standardized curriculum skills
Prevalence denominator: 80

Study Design

Comparative Analysis

Industry skill requirements
vs
University curriculum representation

Analytical Approach

Skill Standardization → Skill Matching → Eight Analyses → Alignment, Gap & Coverage Findings

Skill Standardization Framework

1 Skill Extraction

- 130 Industry Job Description
- 80 University Curricula
- Skills extracted independently from both datasets

2 Skill Vocabulary Standardization

- Synonyms & abbreviations consolidated
- Equivalent terms mapped to one common skill label
- Example: NLP → Natural Language Processing
- Cloud variants → Cloud Platforms (AWS / Azure / GCP)

3 Skill Matching

Exact Match Same skill expression
Normalized-Match Different wording, same standardized skill
Conceptual-Match Broader/related curriculum representation
No Match No corresponding curriculum skill

4 Match Status

- **Covered** → Direct representation
- **Partially Covered** → Broader/conceptual representation
- **Industry Only** → No curriculum counterpart representation



ROLE IN THE STUDY

Skills were standardized to create a common vocabulary and then matched across industry and curriculum datasets, forming the basis for alignment, skill-gap and coverage analysis.

ANALYSIS 1 — SKILL FREQUENCY ANALYSIS

OBJECTIVE

Identify the most frequently represented individual skills in Industry job descriptions and university Curriculum documents.

METHOD / FORMULA

Skill Prevalence = Skill Frequency / Total Documents × 100

Prevalence enables comparison despite different dataset sizes.

Industry — 130 job-description documents

Rank	Skill	Skill Prevalence	Skill Frequency (documents)
1	Python	90.00%	117
2	Communication & Collaboration	84.62%	110
3	Machine Learning	82.31%	107
4	SQL	53.08%	69

Curriculum — 80 curriculum documents

Rank	Skill	Skill Prevalence	Skill Frequency (documents)
1	Statistics	100.00%	80
2	Python	98.75%	79
3	Machine Learning	93.75%	75
4	Data Structures	90.00%	72

KEY INSIGHT

Industry is led by Python, while curriculum is led by Statistics.

Frequency identifies the most prominent individual skills; prevalence provides the fairer comparison across the two datasets.

ANALYSIS 2 — CATEGORY FREQUENCY ANALYSIS

OBJECTIVE

Compare the relative emphasis placed on broader skill categories across Industry and Curriculum datasets.

METHOD / FORMULA

Category Contribution (%) =

Category Frequency / Total Category Frequency × 100

Measures the relative emphasis of each skill category within each dataset.

Skill Category	Industry	Curriculum
Machine learning	16.00	10.29
Generative AI and NLP	15.64	4.25
Professional and soft skills	15.50	0.54
Data Analysis and Statistics	10.71	25.66

KEY INSIGHT

Industry and curriculum place different levels of emphasis across skill categories.

Industry emphasis is strongest in Machine Learning, Generative AI & NLP, and Professional & Soft Skills, while curriculum emphasis is highest in Data Analysis & Statistics and Software Engineering.

ANALYSIS 3 — PREVALENCE ALIGNMENT ANALYSIS

OBJECTIVE

Measure how closely Industry and Curriculum skill prevalence align across comparable standardized skills.

METHOD / FORMULA

Alignment = Min(Industry Prevalence, Curriculum Prevalence) / Max (Industry Prevalence, Curriculum Prevalence)

Percentage-Point Difference = Industry Prevalence − Curriculum Prevalence

Standardized Skill	Industry Prevalence	Curriculum Prevalence	Alignment
Feature Engineering	23.85%	23.75%	99.58%
Python	90.00%	98.75%	91.14%
Machine Learning	82.31%	93.75%	87.80%
SQL	53.08%	40.00%	75.36%

KEY INSIGHT

Alignment varies substantially across comparable skills.

Core skills such as Feature Engineering, Python and Machine Learning show relatively strong prevalence alignment, while Cloud Platforms and Causal Inference show substantial divergence.

ANALYSIS 4 — SKILL GAP ANALYSIS

OBJECTIVE

Identify the direction and magnitude of prevalence differences between Industry and Curriculum for comparable skills.

METHOD / FORMULA

Skill Gap = Industry Prevalence − Curriculum Prevalence

Absolute Gap = |Skill Gap|

Positive = Industry Higher , Negative = Curriculum Higher

Standardized Skill	Industry	Curriculum	Skill Gap
Cloud Platforms	48.46%	16.25%	+32.21 pp
Generative AI	34.62%	10.00%	+24.62 pp
Statistical Analysis	47.69%	100.00%	−52.31 pp
Deep Learning	33.85%	83.75%	−49.90 pp

Largest Industry Gap

Cloud Platforms

+32.21 pp

Largest Curriculum Gap

Causal Inference

−98.46 pp

KEY INSIGHT

The largest gaps are not uniform in direction. Industry prevalence is higher for skills such as Cloud Platforms and Generative AI, while curriculum prevalence is higher for several foundational and conceptual areas. These curriculum-higher results should be interpreted as evidence of strong foundational coverage, not as a lack of value of those subjects.

ANALYSIS 5 — CURRICULUM COVERAGE ANALYSIS

OBJECTIVE

Assess how extensively industry-identified skills are represented in the Curriculum, including exact and conceptual coverage.

METHOD / FORMULA

Strict CCR = Covered Skills / Total Industry Skills × 100

Broad CCR = (Covered + Partially Covered) / Total Industry Skills × 100

TOTAL INDUSTRY SKILLS 81

STRICT COVERAGE 30.86%
(25 Covered)

BROAD COVERAGE 58.02%
(25 Covered + 22 Partially Covered)

Priority Tier	Total Skills	Covered	Partial	Industry Only
Tier 1 ≥50	5	3	0	2
Tier 2 25–49	10	6	4	0
Tier 3 10–24	9	1	3	5
Tier 4 <10	57	15	15	27

KEY INSIGHT

Curriculum coverage is stronger for higher-priority industry skills, but 34 of 81 industry skills have no curriculum counterpart. Strict coverage is 30.86%, while Broad coverage rises to 58.02% when conceptual matches are included. Tier 1 and Tier 2 each achieve 60% strict coverage.

ANALYSIS 6 — EMERGING TECHNOLOGY ALIGNMENT

OBJECTIVE

Evaluate how effectively the curriculum represents selected emerging technologies identified in industry requirements.

METHOD / FORMULA

ETAR = Covered Emerging Technologies / Total Selected Technologies × 100

Broad ETAR = (Covered + Partially Covered) / Total Selected Technologies × 100

14 technologies selected from the industry skill inventory based on industry representation and contemporary relevance.

TECHNOLOGIES EVALUATED 14

STRICT ETAR 14.29%
(2 Covered)

BROAD COVERAGE 28.57%
(2 Covered + 2 Partially Covered)

Status	Technologies	Share
Covered	2	14.29%
Partially Covered	2	14.29%
Industry Only	10	71.43%

KEY INSIGHT

Most selected emerging technologies have no identified curriculum counterpart. Only 2 of 14 technologies are strictly covered, while 10 are Industry Only. Broad alignment rises to 28.57% when conceptual coverage is included.

ANALYSIS 7 - INDUSTRY-CURRICULUM RANK DIFFERENCE

OBJECTIVE

Compare the relative priority ranking of matched skills between industry requirements and university curricula.

METHOD / FORMULA

Rank Difference = Industry Rank − Curriculum Rank

Absolute Rank Difference = |Rank Difference|

MATCHED SKILLS	47	
CURRICULUM LAGS	16	34.04%
CURRICULUM LEADS	29	61.70%
ALIGNED	2	4.26%

Measure	Result	Measure
Average Absolute Rank Difference	24.60	Average Absolute Rank Difference
Maximum Absolute Rank Difference	67	Maximum Absolute Rank Difference
Maximum Gap Skill	Predictive Analytics	Maximum Gap Skill

KEY INSIGHT

Skill presence does not guarantee equal priority. Only 2 of 47 matched skills have identical ranks, while the average absolute rank difference is 24.60. Generative AI and Cloud Platforms show substantial industry-priority gaps.

ANALYSIS 8 — CHI-SQUARE & CRAMÉR'S V

OBJECTIVE

Examine whether skill category and curriculum coverage status are associated.

METHOD / FORMULA

χ^2 = Σ (Observed − Expected)² / Expected

Expected = (Row Total × Column Total) / Grand Total

Cramér's V = $\sqrt{[\chi^2 / (N \times \min(r-1, c-1))]}$

CHI-SQUARE	$\chi^2 = 61.65$
DEGREES OF FREEDOM	df = 26
CRAMÉR'S V	0.617
P-VALUE	0.0001002

Coverage Status	Skills	Share
Covered	25	30.86%
Partially Covered	22	27.16%
Industry Only	34	41.98%

KEY INSIGHT

Coverage patterns vary meaningfully across skill categories, but the statistical result should be interpreted cautiously because the expected-frequency assumption is substantially violated.

CROSS-ANALYSIS SYNTHESIS

OBJECTIVE

Integrate findings across the eight analyses to identify the overall pattern of Industry–Curriculum alignment.

Dimension	What the analyses show
Skill Presence	Core skills such as Python and Machine Learning are strongly represented in both datasets.
Skill Priority	Industry and curriculum often prioritize the same skills differently, with an average absolute rank difference of 24.60 across 47 matched skills.
Skill Prevalence	Overall prevalence alignment is 33.18%, indicating substantial variation across comparable skills.
Curriculum Coverage	30.86% strict coverage rises to 58.02% broad coverage when conceptual matches are included.

OVERALL RESEARCH INSIGHT

The curriculum provides a strong foundation in core Data Science skills, but alignment becomes uneven for applied, emerging, deployment-oriented and professional competencies. Industry–Curriculum gaps are therefore not limited to whether a skill is present, but also involve how strongly it is represented and prioritized.

KEY FINDINGS

Strong Core Foundation

Core skills are strongly represented
Python and Machine Learning show high representation in both Industry and Curriculum, indicating a strong foundation in core Data Science competencies.

Uneven Prevalence Alignment

Overall alignment varies substantially
Average prevalence alignment across 47 comparable skills is 33.18%, with strong alignment for some skills and substantial divergence for others.

Coverage Gap

Curriculum coverage is incomplete
25 of 81 industry skills have strict curriculum coverage, while 22 additional skills have partial or conceptual coverage.

Priority Mismatch

Skill presence does not guarantee equal priority
Among 47 matched skills with comparable ranks, only 2 have identical ranks, with an average absolute rank difference of 24.60.

Emerging Technology Gap

Selected Emerging technologies show limited direct coverage
Only 2 of 14 selected emerging technologies have direct curriculum coverage, while 10 are Industry Only.

CURRICULUM IMPLICATIONS

Objective Translate the observed Industry–Curriculum gaps into areas that may warrant curriculum review and development.

Strengthen Applied & Industry-Oriented Skills

Greater attention may be given to skills showing strong industry prevalence but lower curriculum representation.

Integrate Emerging Technologies

Selected emerging technologies with limited or no curriculum representation may warrant consideration within relevant courses or practical modules.

Review Skill Priority, Not Only Presence

Curriculum review should consider how strongly skills are prioritized, not simply whether they appear in the curriculum.

Implication for Curriculum Design The findings support a balanced curriculum approach: retain strong foundational competencies while strengthening applied, emerging, deployment-oriented and professional competencies where industry evidence indicates gaps.

These implications identify areas for curriculum review; they do not imply that every Industry Only skill should be added to every curriculum.

LIMITATIONS OF THE STUDY

Dataset Scope

The analysis is based on 130 industry job descriptions and 80 university curricula and therefore reflects the selected datasets rather than the entire Data Science job market or all university programs.

Skill Standardization

Skill matching involves standardized terminology and conceptual relationships. Some matches therefore require methodological judgement rather than purely exact textual matching.

Emerging Technology Scope

The emerging-technology analysis focuses on 14 selected technologies from the industry skill inventory and should not be interpreted as an exhaustive assessment of all emerging technologies.

Frequency ≠ Importance

Skill frequency reflects how often a skill appears across documents. It does not directly measure educational importance, difficulty, learning depth, or employer preference.

Exploratory Statistical Analysis

The Chi-square analysis is exploratory because 40 of 42 expected cells (95.24%) were below 5. Therefore, the p-value should not be treated as definitive statistical evidence.

The study identifies patterns of Industry–Curriculum alignment and skill gaps; it does not establish causality or prove that curriculum differences directly cause employment outcomes.

Conclusion

The study finds strong alignment in foundational Data Science skills, but uneven alignment in applied, emerging, deployment-oriented and professional competencies.

Curriculum relevance depends not only on whether a skill is present, but also on how strongly it is represented, prioritized and connected to industry requirements.

Thank You | Questions & Discussion